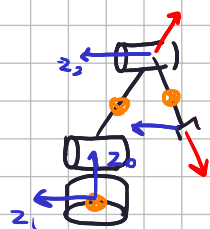


Table 1 contains the Denavit-Hartenberg parameters of a robot with three revolute joints.

| $i$ | $\alpha_i$ | $a_i$     | $d_i$ | $\theta_i$ |
|-----|------------|-----------|-------|------------|
| 1   | $\pi/2$    | 0         | 0     | $\theta_1$ |
| 2   | 0          | $a_2 > 0$ | 0     | $\theta_2$ |
| 3   | 0          | $a_3 > 0$ | 0     | $\theta_3$ |

Use the recursive algorithm based on moving frames to compute the kinetic energy of this robot, making reasonable assumptions to simplify the inertial properties of the links, e.g., assume that each link has a cylindric body with uniformly distributed mass and center of mass on the kinematic axis of the link. Provide at the end the robot inertia matrix  $M(q)$  and determine a minimal linear parametrization of the inertial term  $M(q)\ddot{q}$ .

The robot is:



$${}^0T_1 = \begin{pmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^1T_2 = \begin{pmatrix} C_2 & -S_2 & 0 & a_2 C_2 \\ S_2 & C_2 & 0 & a_2 S_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^2T_3 = \begin{pmatrix} C_3 & -S_3 & 0 & a_3 C_3 \\ S_3 & C_3 & 0 & a_3 S_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^1w_1 = {}^0R_1^T (0 \ 0 \ \dot{q}_1)^T = \begin{pmatrix} C_1 & S_1 & 0 \\ 0 & 0 & 1 \\ S_1 & -C_1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \dot{q}_1 \end{pmatrix} = \begin{pmatrix} 0 \\ \dot{q}_1 \\ 0 \end{pmatrix}$$

$${}^1p_{c1} = 0$$

$$\Rightarrow {}^1v_1 = {}^0R_1^T \left[ (0 \ 0 \ \dot{q}_1)^T \times (0 \ 0 \ 0)^T \right] = 0$$

$${}^2p_{c2} = (d_2 \ 0 \ 0)^T$$

$$v_{c1} = 0 + (0 \ \dot{q}_1 \ 0) \times {}^1p_{c1} = 0$$

$${}^3p_{c3} = (d_3 \ 0 \ 0)^T$$

$${}^2w_2 = \begin{pmatrix} C_2 & S_2 & 0 \\ -S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ \dot{q}_1 \\ \dot{q}_2 \end{pmatrix} = \begin{pmatrix} S_2 \dot{q}_1 \\ C_2 \dot{q}_1 \\ \dot{q}_2 \end{pmatrix}$$

$${}^2v_2 = \begin{pmatrix} C_2 & S_2 & 0 \\ -S_2 & C_2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -\dot{q}_2 & \dot{q}_1 \\ \dot{q}_2 & 0 & 0 \\ -\dot{q}_1 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_2 C_2 \\ a_2 S_2 \\ 0 \end{pmatrix} = \begin{cases} 0 \\ a_2 \dot{q}_2 \\ -\dot{q}_1 a_2 C_2 \end{cases}$$

$$v_{c2} = \begin{pmatrix} 0 \\ a_2 \dot{q}_2 \\ -\dot{q}_1 a_2 C_2 \end{pmatrix} + \begin{pmatrix} 0 & -\dot{q}_1 & \dot{q}_1 C_2 \\ \dot{q}_2 & 0 & -\dot{q}_1 S_2 \\ -\dot{q}_1 C_2 & \dot{q}_1 S_2 & 0 \end{pmatrix} \begin{pmatrix} -d_2 \\ 0 \\ 0 \end{pmatrix} = \begin{cases} 0 \\ \dot{q}_2 (a_2 - d_2) \\ \dot{q}_1 C_2 (d_2 - a_2) \end{cases}$$

$${}^3w_3 = \begin{pmatrix} C_3 & S_3 & 0 \\ -S_3 & C_3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} S_2 \dot{q}_1 \\ C_2 \dot{q}_1 \\ \dot{q}_2 + \dot{q}_3 \end{pmatrix} = (\dot{q}_1 S_{23} \quad \dot{q}_1 C_{23} \quad \dot{q}_2 + \dot{q}_3)^T$$

$${}^3v_3 = \begin{pmatrix} C_3 & S_3 & 0 \\ -S_3 & C_3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \left[ \begin{pmatrix} 0 \\ a_2 \dot{q}_2 \\ -\dot{q}_1 a_2 C_2 \end{pmatrix} + \begin{pmatrix} 0 & -(\dot{q}_2 + \dot{q}_3) & \dot{q}_1 C_2 \\ \dot{q}_2 + \dot{q}_3 & 0 & -\dot{q}_1 S_2 \\ -\dot{q}_1 C_2 & \dot{q}_1 S_2 & 0 \end{pmatrix} \begin{pmatrix} a_3 C_3 \\ a_3 S_3 \\ 0 \end{pmatrix} \right]$$

$$= \begin{pmatrix} C_3 & S_3 & 0 \\ -S_3 & C_3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -a_3 S_3 (\dot{q}_2 + \dot{q}_3) \\ a_3 C_3 (\dot{q}_2 + \dot{q}_3) + a_2 \dot{q}_2 \\ -\dot{q}_1 a_3 C_{23} - a_2 C_2 \dot{q}_1 \end{pmatrix} = \begin{cases} a_2 S_3 \dot{q}_2 \\ a_3 (\dot{q}_2 + \dot{q}_3) + a_2 C_3 \dot{q}_2 \\ -\dot{q}_1 (a_3 C_{23} + a_2 C_2) \end{cases}$$

$$v_{c3} = {}^3v_3 + \begin{pmatrix} 0 & -(\dot{q}_2 + \dot{q}_3) & \dot{q}_1 C_{23} \\ \dot{q}_2 + \dot{q}_3 & 0 & -\dot{q}_1 S_{23} \\ -\dot{q}_1 C_{23} & \dot{q}_1 S_{23} & 0 \end{pmatrix} \begin{pmatrix} -d_3 \\ 0 \\ 0 \end{pmatrix} = \begin{cases} a_2 S_3 \dot{q}_2 \\ -d_3 (\dot{q}_2 + \dot{q}_3) + a_3 (\dot{q}_2 + \dot{q}_3) + a_2 C_3 \dot{q}_2 \\ -\dot{q}_1 (a_3 C_{23} + a_2 C_2 - d_3 C_{23}) \end{cases}$$

$$T_1 = \frac{1}{2} I_{1y} \dot{q}_1^2$$

$$T_2 = \frac{1}{2} \left( I_{2x} S_2^2 \dot{q}_1^2 + I_{2y} C_2^2 \dot{q}_1^2 + I_{2z} \dot{q}_2^2 \right) + \frac{1}{2} m_2 \left[ \dot{q}_2^2 (a_2 - d_2)^2 + \dot{q}_1^2 C_2^2 (d_2 - a_2)^2 \right]$$

$$T_3 = \frac{1}{2} \left[ I_{3x} S_{23}^2 \dot{q}_1^2 + I_{3y} C_{23}^2 \dot{q}_1^2 + I_{3z} (\dot{q}_2^2 + \dot{q}_3^2 + 2\dot{q}_2 \dot{q}_3) \right] + \frac{1}{2} m_3 \left[ 2a_2^2 \dot{q}_2^2 + (a_3 - d_3)^2 (\dot{q}_2^2 + \dot{q}_3^2 + 2\dot{q}_2 \dot{q}_3) + 2a_2 (a_3 - d_3) C_3 (\dot{q}_2^2 + \dot{q}_2 \dot{q}_3) + \dot{q}_1^2 \left[ C_{23}^2 (a_3 - d_3)^2 + a_2^2 C_2^2 + 2a_2 (a_3 - d_3) C_2 C_{23} \right] \right]$$

$$M_{11} = I_{1y} + I_{2x} + (I_{2y} - I_{2x}) C_2^2 + m_2 C_2^2 (d_2 - a_2)^2 + I_{3x} + (I_{3y} - I_{3x}) C_{23}^2 + m_3 \left[ C_{23}^2 (a_3 - d_3)^2 + a_2^2 C_2^2 + 2a_2 (a_3 - d_3) C_2 C_{23} \right]$$

$$= C_2^2 \left[ I_{2y} - I_{2x} + m_2 (d_2 - a_2)^2 + m_3 a_2^2 \right]^{e_2} + C_{23}^2 \left[ I_{3y} - I_{3x} + m_3 (a_3 - d_3)^2 \right]^{e_3} + e_1 \left[ I_{1y} + I_{2x} + I_{3x} + 2m_3 a_2 (a_3 - d_3) C_2 C_{23} \right]$$

$M_{11} = e_1 + e_2 C_2^2 + e_3 C_{23}^2 + 2e_4 C_2 C_{23}$

$$M_{22} = I_{22} + m_2 (a_2 - d_2)^2 + I_{32} + m_3 (a_2^2 + (a_3 - d_3)^2) + m_3 a_2 (a_3 - d_3) C_3$$

$$= e_4 C_3 + e_5$$

$$M_{23} = I_{32} + m_3 (a_3 - d_3)^2 + \frac{1}{2} m_3 a_2 (a_3 - d_3) C_3 = \frac{1}{2} e_4 C_3 + e_6$$

$$M_{33} = I_{32} + m_3 (a_3 - d_3)^2 = e_6$$

The expanded inertial term  $M(q)\ddot{q}$  is:

$$\begin{cases} (e_1 + e_2 C_2^2 + e_3 C_{23}^2 + 2e_4 C_2 C_{23}) \ddot{q}_1 \\ (e_4 C_3 + e_5) \ddot{q}_2 + (\frac{1}{2} e_4 C_3 + e_6) \ddot{q}_3 \\ (\frac{1}{2} e_4 C_3 + e_6) \ddot{q}_2 + e_6 \ddot{q}_3 \end{cases} \Rightarrow \text{transform it in } \gamma e$$

$$\begin{bmatrix} \ddot{q}_1 & \ddot{q}_1 C_2^2 & \ddot{q}_1 C_{23}^2 & 2C_2 C_{23} \ddot{q}_1 & 0 & 0 \\ 0 & 0 & 0 & \ddot{q}_2 C_3 + \frac{1}{2} C_3 \ddot{q}_3 & \ddot{q}_2 & \ddot{q}_3 \\ 0 & 0 & 0 & \frac{1}{2} C_3 \ddot{q}_2 & 0 & \ddot{q}_2 + \ddot{q}_3 \end{bmatrix} e = \begin{pmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \end{pmatrix} \left. \begin{matrix} e_1 \\ e_2 \\ e_3 \end{matrix} \right\} \text{dep.}$$

lin dep.

$$\underbrace{\begin{bmatrix} \ddot{q}_1 & 2C_2C_3\ddot{q}_1 & 0 & 0 \\ 0 & \ddot{q}_2C_3 + \frac{1}{2}C_3\ddot{q}_3 & \ddot{q}_2 & \ddot{q}_3 \\ 0 & \frac{1}{2}C_3\ddot{q}_2 & 0 & \ddot{q}_2 + \ddot{q}_3 \end{bmatrix}}_{Y_{ind}} = \begin{pmatrix} e_1 \\ e_4 \\ e_5 \\ e_6 \end{pmatrix} + \Delta \begin{pmatrix} e_2 \\ e_3 \end{pmatrix}$$

$$\Delta = \begin{pmatrix} C_2^2 & C_{23}^2 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

## Exercise 2

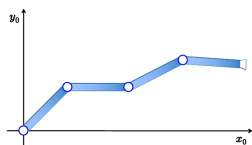


Figure 1: A 4R planar robot.

Consider the 4R planar robot in Fig. 1, having links with unit length. The primary task for the end-effector is to point at a moving target in the plane  $(x_0, y_0)$ . The available extra degrees of freedom of the robot are used to keep the joints close to the middle of their ranges, which are defined, using D-H variables, as

$$q_1 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad q_2 \in \left[0, \frac{\pi}{2}\right] \quad q_i \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right], \quad i = 3, 4. \quad (1)$$

The task of pointing regards the orientation of the e-e :

$$\alpha(q) = \sum_{i=1}^4 q_i \quad \text{hence} \quad J_\alpha = \begin{pmatrix} 1 & 1 & 1 & 1 \end{pmatrix}.$$

The function that measure the distance between the ranges is

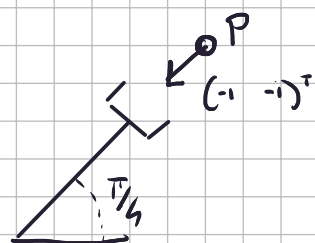
$$G(q) = \frac{1}{2} \left[ \left( \frac{q_1}{\pi} \right)^2 + \left( \frac{q_2 - \pi/4}{\pi/2} \right)^2 + \left( \frac{q_3}{\pi/2} \right)^2 + \left( \frac{q_4}{\pi/2} \right)^2 \right]$$

$$\nabla G = - \begin{pmatrix} \frac{q_1}{\pi^2} & \frac{q_2 - \pi/4}{\pi^2/4} & \frac{q_3}{\pi^2/4} & \frac{q_4}{\pi^2/4} \end{pmatrix}^T$$

$$\Rightarrow \dot{q} = J_\alpha^\# \dot{\alpha} + \underbrace{(\mathbf{I} - J_\alpha^\# J_\alpha)}_P \nabla G$$

$$\text{In } q = \left(0 \quad \frac{\pi}{2} \quad 0 \quad -\frac{\pi}{4}\right) \Rightarrow \alpha = \frac{\pi}{4}$$

If the target is being pointed with  $\alpha = \frac{\pi}{4}$  and it moves at  $\dot{p} = (-1 \ -1)^T$ , we don't need to change our orientation:



so we just consider the preferred velocity.

$$\nabla G = (0 \quad -0.3183 \quad 0 \quad 0.3183)^T$$

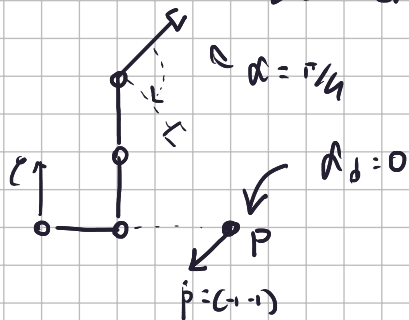
$$J^\# = \frac{1}{4} (1 \quad 1 \quad 1 \quad 1)^T$$

$$\ddot{q} = (I - J^\# J) \nabla G = (0 \quad -0.3183 \quad 0 \quad 0.3183)^T = \nabla G$$

If  $p_t = (0 \quad 2.5)$  we consider the angle displacement:  $\nearrow e = \alpha_d - \alpha$

$$\dot{q} = J_\alpha^\# (\dot{\alpha}_d + \kappa e) + P \nabla G \rightarrow \kappa = 1$$

the error is  $0 - \frac{\pi}{4} = -\frac{\pi}{4}$  so



$$J_\alpha^\# \left(-\frac{\pi}{4}\right) + P \nabla G = -\frac{\pi}{16} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + 0.3183 \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

### Exercise 3

The dynamics of a robot with  $n$  elastic joints moving in the absence of gravity is described by the  $2n$  second-order differential equations

$$M(q)\ddot{q} + c(q, \dot{q}) + K(q - \theta) = 0$$

$$B_m \ddot{\theta} + K(\theta - q) = u,$$

(2)

where  $K_P$  and  $K_D$  are diagonal and positive definite gain matrices, and  $\theta_d \in \mathbb{R}^n$  is a desired constant motor position. Prove that

(3)

$$u = K_P(\theta_d - \theta) - K_D \dot{\theta}, \quad (4)$$

where (2) are the  $n$  link equations and (3) are the  $n$  motor equations, with link positions  $q \in \mathbb{R}^n$  and motor positions  $\theta \in \mathbb{R}^n$  as generalized coordinates.  $K$  is the diagonal, positive definite stiffness matrix of the joints, while  $B_m$  is the diagonal, positive definite inertia matrix of the motors.

For the input torque  $u \in \mathbb{R}^n$ , consider the PD control law on the motor variables

is the unique, globally asymptotically stable equilibrium state for the closed-loop system made by eqs. (2) and (3) under the control law (4).

I choose the following Lyapunov candidate

$$V = \frac{1}{2} \dot{q}^T M \dot{q} + \frac{1}{2} (q - \theta)^T K (q - \theta) + \frac{1}{2} (\theta_d - \theta)^T K_P (\theta_d - \theta) + \frac{1}{2} \dot{\theta}^T B_m \dot{\theta}$$

this function is zero only on  $\theta = \theta_d = q$ ,  $\dot{\theta} = \dot{q} = 0$

The derivative is

$$\dot{V} = \frac{1}{2} \dot{q}^T \dot{M} \dot{q} + \dot{q}^T M \ddot{q} + (q - \theta)^T K (\dot{q} - \dot{\theta}) + (\theta_d - \theta)^T K_P \dot{\theta} + \dot{\theta}^T B_m \ddot{\theta} =$$

$$= \dot{q}^T \left( \frac{1}{2} \dot{M} \dot{q} + C \dot{q} + K(q - \theta) \right) + (q - \theta)^T K (\dot{q} - \dot{\theta}) - (\theta_d - \theta)^T K_P \dot{\theta} + \dot{\theta}^T (K(\theta - q) - K_P(\theta_d - \theta) - K_D \dot{\theta})$$

$$= \dot{q}^T K(q - \theta) + (q - \theta)^T K (\dot{q} - \dot{\theta}) + (\theta_d - \theta)^T K_P \dot{\theta} + \dot{\theta}^T (K(\theta - q) - K_P(\theta_d - \theta) - K_D \dot{\theta}) =$$

$$- \dot{\theta}^T K_D \dot{\theta} + \dot{\theta}^T K(\theta - q) +$$

$$\dot{q}^T K(q - \theta) + (q - \theta)^T K (\dot{q} - \dot{\theta}) = - \dot{\theta}^T K_D \dot{\theta} \leq 0$$

So  $\dot{V} \leq 0$ , we have to use LaSalle.

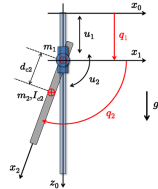
$$\dot{V} = \frac{1}{2} \dot{q}^T \dot{M} \dot{q} + \dot{q}^T M \ddot{q} + (q - \theta)^T K (\dot{q} - \dot{\theta}) + (\theta_d - \theta) K_p \dot{\theta} + \dot{\theta}^T B_m \ddot{\theta} = 0$$

$\Rightarrow \dot{q} = 0 \quad \dot{\theta} = 0 \quad q = \theta = \theta_d$  let's see the dynamics here:

$$\begin{cases} M \ddot{q} = 0 \\ B_m \ddot{\theta} = 0 \end{cases} \Rightarrow \ddot{q} = \ddot{\theta} = 0 \Rightarrow \{(\theta_d, \theta_d, 0, 0)\} \text{ is the largest invariant set.}$$

#### Exercise 4

The PR robot in Fig. 2 may be subject to a generic unknown fault  $u_{f1}$  on the force produced by first actuator. Based on the symbolic terms of the dynamic model of this robot, design a scalar residual function  $r_1(t)$  such that  $r_1(t) \equiv 0$  in the absence of a fault of this actuator, while it evolves otherwise as  $\dot{r}_1(t) = k_1(u_{f1}(t) - r_1(t))$ , for a given  $k_1 > 0$ . If the unknown fault consists in a constant force  $u_{f1} = 2$  N being subtracted to the commanded force  $u_1$  at the first joint starting from the instant  $t_0 = 0$ , what will be the evolution of  $r_1(t)$  for  $t \geq t_0$  and its value at steady state?



We need the model

$$p_{c1} = (0 \ 0 \ q_1)^T \Rightarrow T_1 = \frac{1}{2} m_1 \dot{q}_1^2$$

$$p_{c2} = (-d_{c2} c_2 \ 0 \ d_{c2} s_2 + q_1)^T \Rightarrow T_2 = \frac{1}{2} I_2 \dot{q}_2^2 + m_2 (d_{c2}^2 \dot{q}_2^2 + \dot{q}_1^2 + 2d_{c2} c_2 \dot{q}_1 \dot{q}_2)$$

$$M_{11} = m_1 + m_2 \quad M_{12} = m_2 d_{c2} c_2 \quad M_{22} = I_2 + m_2 d_{c2}^2$$

$$M = \begin{pmatrix} e_1 & e_2 c_2 \\ e_2 c_2 & e_3 \end{pmatrix} \quad U = -g_0 (m_1 q_1 + m_2 q_1 + m_2 d_{c2} s_2)$$

$$g = \begin{cases} -g_0 e_1 \\ -g_0 e_2 c_2 \end{cases} \quad C_1 = \begin{pmatrix} 0 & 0 \\ 0 & -e_2 s_2 \end{pmatrix} \Rightarrow c_1 = -e_2 s_2 \dot{q}_2^2 \quad c_2 = 0$$

We compute

$$\alpha_1 = -g_0 e_1 \quad \alpha_2 = -\frac{1}{2} \dot{q}^T \begin{pmatrix} 0 & -e_2 s_2 \\ -e_2 s_2 & 0 \end{pmatrix} \dot{q} - g_0 e_2 c_2 = -2e_2 s_2 \dot{q}_1 \dot{q}_2 - g_0 e_2 c_2$$

We choose  $K = \begin{pmatrix} \kappa_1 \\ \kappa_2 \end{pmatrix}$  and set

$$r = K \left( \int (\tau - \alpha - r) dt - p \right) \quad \text{with} \quad p = M \dot{q} = \begin{cases} e_1 \dot{q}_1 + e_2 c_2 \dot{q}_2 \\ e_2 c_2 \dot{q}_1 + e_3 \dot{q}_2 \end{cases}$$

this provide the behaviour:  $\dot{r} = K(\tau - \alpha - r) \Rightarrow \dot{r}_1 = \kappa_1 (u_{s1} - r_1)$ .

Let  $u_{s1} = 2$  so

$$\dot{r}_1 = 2\kappa_1 - r_1 \quad r_1 = 2\kappa_1 - \dot{r}_1 \Rightarrow r_1(t) = 2 \exp(-\kappa_1 t)$$

$$\lim_{t \rightarrow \infty} r_1 = 2$$

